Harry Chandra Tsjan

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Education

Hong Kong University of Science and Technology, MSc. Information Technology — Sep 2026 – July 2027

University Tunku Abdul Rahman, B.Sc. in Software Engineering — Oct 2021 – Oct 2024

Experience

AI Engineer, Carbon AI — Dec 2025 – Aug 2026

- Collaborated with product stakeholders to understand business requirements and translate them into practical AI applications, strengthening cross-functional communication and problem-solving.
- Designed and deployed AI-powered workflows using Python, FastAPI, LangChain, and LangGraph to support business use cases and improve information-processing efficiency.
- Developed a hybrid information-retrieval workflow using pgvector, Vertex AI embeddings, and semantic retrieval techniques, improving document retrieval accuracy by approximately 25%
- Evaluated AI-generated outputs and identified failure cases to improve response quality, applying critical judgment when working with AI-generated information.
- Managed production AI services on Google Cloud Platform, adapting to changing technical requirements while supporting reliable delivery of business applications.

Data Science Intern, RosaryLabs — Mar 2025 – Aug 2025

- Designed an MCP-based search and agent workflow enabling dynamic tool selection and multi-turn reasoning for complex information-retrieval task
- Built evaluation pipelines using DeepEval and custom LLM metrics to assess response quality, identify failure cases, and support data-driven workflow improvements.
- Evaluated Qdrant vector search and sparse embedding approaches for Retrieval-Augmented Generation, comparing retrieval strategies for relevance and search quality
- Worked with emerging AI technologies and adapted quickly to new tools, frameworks, and technical requirements.

Software Engineer Intern, Consolsys Sdn Bhd — Oct 2024 – Jan 2025

- Integrated Stimulsoft Reports into the tellering system using Angular and C#, delivering dynamic reporting capabilities and enhanced user experience.
- Designed and implemented transaction data workflows to ensure accurate request routing and reliable journaling of completed transactions within the local database.
- Diagnosed and resolved application issues using debugging techniques, including breakpoints and runtime analysis, to identify root causes and implement long-term solutions that improved system stability.

Projects

NeuroGrade | Python, LangChain, LLM, React, FastAPI

- Developed an AI-assisted application that uses CLIP-based image embeddings to represent brain-imaging cases and identify comparable historical cases based on image characteristics.
- Implemented vector similarity search using Pinecone to retrieve relevant historical cases from an image-embedding database, enabling efficient access to related information.
- Built an AI workflow that combines retrieved evidence with LLM-generated findings into a structured view, allowing users to review information before making their own assessment.
- Automated structured report generation by combining retrieved evidence, AI-assisted findings, and user assessment, streamlining information-processing workflows while maintaining human oversight.

Technologies

Programming: Python, SQL, NoSQL, C#, Java, JavaScript, Dart, Spark

Technologies: MSSQL, PostgreSQL, AWS EC2, Git, Azure Data Studio, Tableau, Snowflake, Docker, Azure, Airflow, dbt

Languages: Cantonese (Basic), English (Native), Mandarin (Native), Malay (Fluent)

Awards & Certifications

AWS Cloud Practitioner Foundational Dec 2024

NVIDIA Fundamentals of Deep Learning Dec 2024

First Runner-Up, ASEAN AI Malaysia Summit Buildathon Oct 2025